Maya Struzak

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Education

Carnegie Mellon University — *Pittsburgh, PA*
PhD in Civil and Environmental Engineering — *Aug 2024 – present*
- Advised by David Rounce and Sarah Fakhreddine

University of Portland — *Portland, OR*
BS in Civil Engineering — *Aug 2020 – May 2024*

INSA Lyon — *Lyon, France*
Exchange in Civil Engineering and Urbanism — *Spring 2023*

Experience

Graduate Research Assistant — *Pittsburgh, PA*
Carnegie Mellon University — *Jan 2024 – present*
- Collaborating with the Chesapeake Bay Program, The Water Institute, and RAND to evaluate BMP effectiveness under future climate scenarios
- Modeling pollutant removal efficiencies for nitrogen, phosphorus, and sediment across four physiographic regions using the APEX model
- Developing Python tools for automated data processing, analysis, and visualization of multi-regional watershed model outputs

Civil Engineering Production Intern — *Portland, OR*
Kimley-Horn — *June 2023 – Aug 2023*
- Drafted and modeled engineering designs using Civil 3D and ArcGIS
- Sought out mentorship and training from senior engineers

International Student Researcher — *Lyon, France*
INSA Lyon — *Mar 2023 – June 2023*
- Completed independent research thesis on modeling and assessing bioretention cell resilience to climate change
- Conducted literature review of 86 publications and curated case studies to validate bioretention models
- Collaborated with PhD candidate on preliminary model validation for CANOE Hydrobox

Undergraduate Research Assistant — *Portland, OR*
University of Portland — *Mar 2022 – Dec 2022*
- Studied pollutant removal efficiency of biochar in bioretention systems
- Designed and conducted experimental trials using pilot-scale bioretention systems and performed wet lab analysis of stormwater effluent
- Prepared a manuscript as first author for Journal of Environmental Engineering

Undergraduate Research Assistant — *Portland, OR*
Portland State University — *Sept 2022 – Dec 2022*
- Conducted field sample collection, preparation, and documentation of moss samples from urban trees
- Performed preliminary microplastics analysis

Projects

Physics-Informed Neural Networks for Water Quality *Carnegie Mellon Univ.*
Mar 2026 – present
- Developed satellite imagery models to predict chlorophyll-a content using Sentinel-2 data
- Compared traditional band-ratio algorithms with physics-constrained neural networks

Assessing Efficiency of Ocean Alkalinity Enhancement (OAE) *NASA JPL*
June 2025 – Aug 2025
- Conducted experiments using a 1D model to assess the efficiency of OAE along different latitudes and dosing schedules

Representative Characteristics of the Chesapeake Bay Watershed *Carnegie Mellon Univ.*
Mar 2025 – May 2025
- Conducted multi-parameter characterization of four physiographic regions using satellite data, soil databases, and groundwater measurements

Outfall 65 Stormwater Treatment *University of Portland*
Sept 2023 – May 2024
- Designed stormwater treatment facility integrating green infrastructure with treatment vault for socio-economically vulnerable community

Ocean Power and Energy Burden *Reykjavik University*
June 2022 – Aug 2022
- Compiled review of ocean power technologies; assessed feasibility for reducing energy burden in marginalized coastal communities

Publications

Evaluation of Biochar on Metals, Nutrient, and Microplastics Removal in Bioretention Systems Apr 2024
Maya Struzak, Cara Poor, Jordyn Wolfand, Abigail Radke
10.1061/JOEEDU.EENG-7487 (Journal of Environmental Engineering)

Presentations

Agricultural BMP Performance Under Climate Change *Pittsburgh, PA*
National Adaptation Forum *Spring 2026*

Impact of Future Hydrology on BMPs in the Chesapeake Bay Watershed *Pittsburgh, PA*
Spring Industry Workshop: Building Resilient Infrastructure *Spring 2025*

Sustainable Stormwater Design for Outfall 65 *Portland, OR*
Shiley School of Engineering Capstone Showcase *Spring 2024*

Equitable Engineering in Cross Cultural Settings *Portland, OR*
University of Portland Founder's Day *Spring 2023*
- Capstone presentation for NAE Grand Challenge Scholars Program (see "Awards")

Efficiency of Biochar-Amended Bioretention Systems *Portland, OR*
EWRG Stormwater Symposium Poster Session *Fall 2023*
- Won distinction "Audience Favorite" (see "Awards")

Awards

Dean's List *University of Portland*
Maintained GPA of over 3.5 *2020-2024*

President's Scholarship *University of Portland*
Merit scholarship of $100,000 towards tuition *2020-2024*

Grand Challenge Scholar *University of Portland*
National Academies of Engineering interdisciplinary honors program for undergraduates *Spring 2024*

Sustainable Stormwater Symposium "Audience Favorite" *Portland, OR*
Won $1000 prize for "Audience Favorite" during poster session *Fall 2023*

- work published in Journal of Environmental Engineering (see "Publications" and "Presentations")

Professional Development

Decision Making Under Deep Uncertainty Summer School *Cambridge, UK*
DMDU Society and RAND *Summer 2026*

NASA Summer School on Satellite Observations and Climate Models *Pasadena, CA*
NASA JPL Center for Climate Science and Keck Institute for Space Studies *Summer 2025*

Renewable Energy Innovation & Sustainability *Reykjavik, Iceland*
Reykjavik University and The Green Program *Summer 2022*

Skills

Programming: Python (geopandas, rasterio, SALib, TensorFlow), R, Git, MATLAB

GIS & Modeling: ArcGIS Pro, Google Earth Engine, APEX, SWMM, AutoCAD Civil3D

Certifications: Engineer in Training (California), AutoCAD Certified User

Languages: English (Native), French (Native), Polish (Intermediate), Spanish (Elementary)

Extracurriculars

- **Environment & Water Resources Institute (EWRI) Graduate Student Chapter**: Assist the president in planning monthly events
- **Graduate Student Outreach Committee:** Plan outreach events promoting STEM education and lead workshops for K-12 students (2024–present)
- **Society of Women Engineers:** Provide mentorship to young women in engineering and lead water resource activities (2022–present)
- **Forest Park Trail Conservation**: Collaborated with non-profit to lead student trail cleanup groups (2021–2024)
- **Public Health Volunteer:** Fundraised and constructed anaerobic digestor toilet systems with local engineers in Anomabo, Ghana (2022)